



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2023

Paper: Sampling Techniques (Theory)

(Common with BS-AD 5th)

Course Code: STAT-305

Roll No.

Time: 3 Hrs.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- Q1) Write down the advantages and disadvantages of sampling?
- Q2) Define standard error and its uses in sampling theory.
- Q3) Explain: Stratified sampling, Proportional allocation, Sampling distribution
- Q4) Write a note on inverse sampling.
- Q5) What are the points to be considered while estimating the sample size?
- Q6) Write a note on circular systematic sampling.

Answer the following questions.

(3x10=30)

Q.2. Q. For simple random sampling, show that the sample mean, $\bar{y}$ is an unbiased estimator of the population mean $\bar{Y}$ and variance of sample mean is given by

$$V(\bar{y}) = \left(\frac{N-1}{Nn} \right) S^2$$

Use Cornfield approach to prove this.

Q.3. If the terms in $\frac{1}{N_h}$ are ignored relative to unity, show that for estimated mean from stratified random

sample of size $n_h \cdot f_h \neq 114$:-

$$V_{opt} \leq V_{prop} \leq V_{ran}$$

Where the optimum allocation is for fixed 'n' and n_h is proportional to $N_h S_h$ ✓

Q.4 Show that the mean of a systematic sample is more precise than the mean of a simple random sample if and only if $S_{sys}^2 > S^2$